605.486 – Android Development

Module 12a

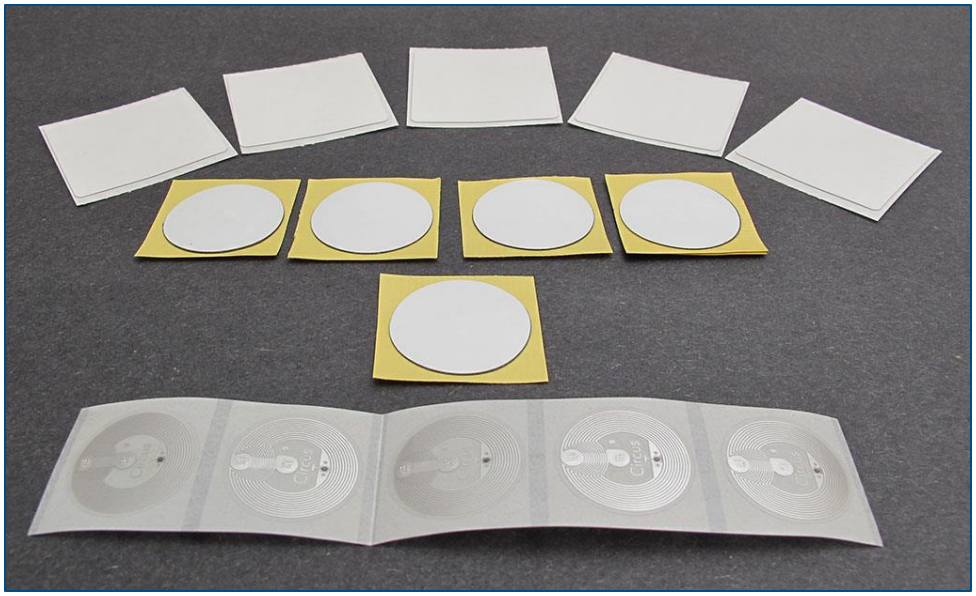
Near-Field Communication (NFC)

Near Field Communication (NFC)

- Short range (10cm max; 4cm practical)
- Tags
- Peer-to-peer with two powered devices
- Not all phones/tablets have NFC chips

NFC Tags

- Usually passive – powered by initiating device
- Usually read-only
- Can be read/write



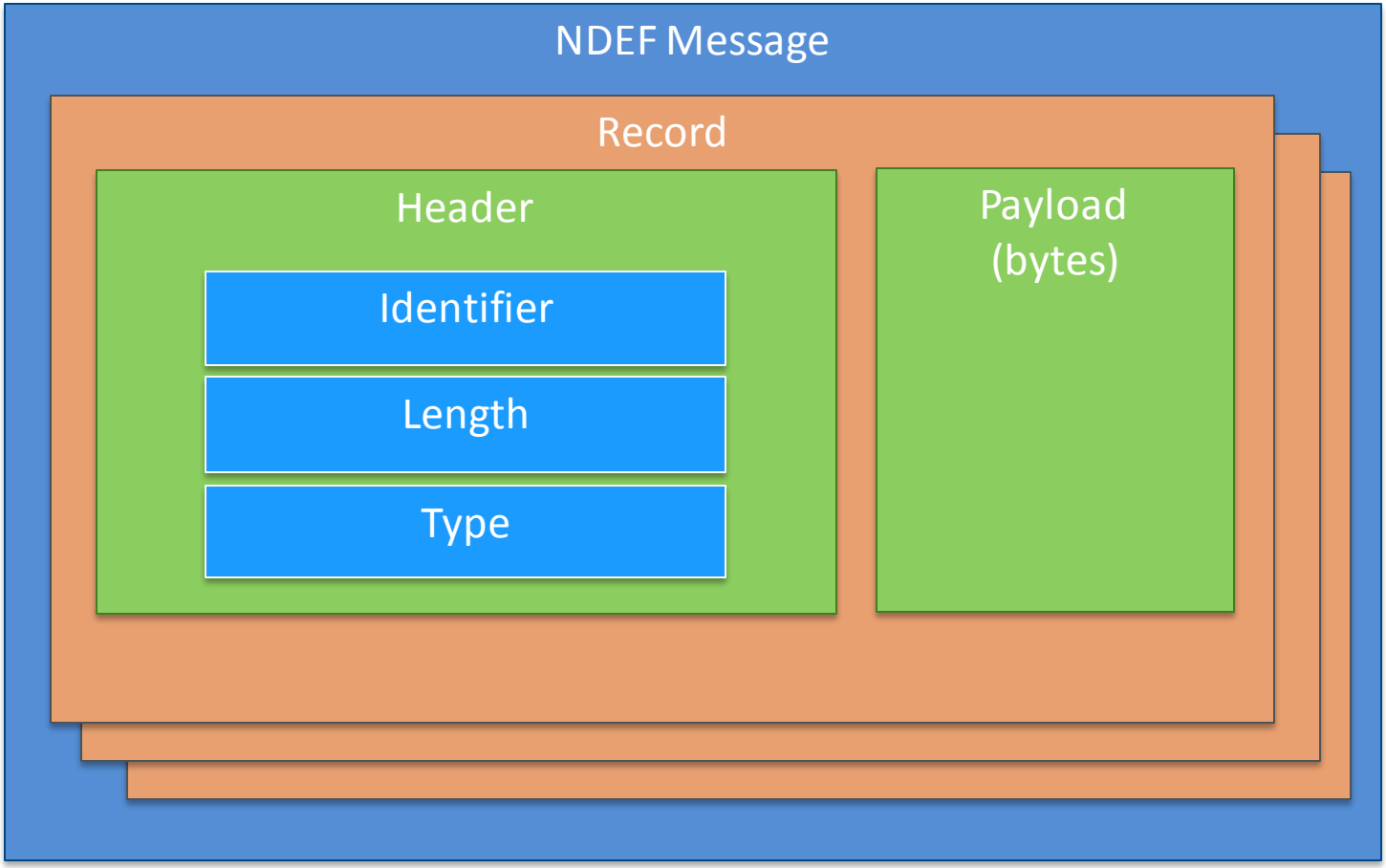
NFC Security

- Short Range
- Device must be unlocked and awake
- Apps must be installed to accept or write data
- Good apps will ask for user confirmation

NFC - Common Uses

- Identification
  - Entry gates/keycards, payments
- Information
  - Web site URL, Contact information, Text
- Bluetooth or other networking setup
- Starting/finding applications
- Send data between phones

NFC Data Exchange Format (NDEF)



Processing Tags on Android

- Framework handles reading data from tag
- Creates Intent based on data
  - URI
  - Mime Type
- Sends Intent with Action

NFC Intents

- **ACTION_NDEF_DISCOVERED**
  - Tag with NDEF encoding read
- ACTION_TECH_DISCOVERED
  - Tag with other encoding
    (or no NDEF_DISCOVERED matches)
- ACTION_TAG_DISCOVERED
  - Tag with other encoding read
    (or no other DISCOVERED matches)

Simple Read – Intent Filter

- NO PERMISSIONS NEEDED
- Register Intent Filter
  - Remember Category DEFAULT!

```
<intent-filter>
    <action android:name="android.nfc.action.NDEF_DISCOVERED" />
    <category android:name="android.intent.category.DEFAULT" />
    <data android:mimeType="text/plain"/>
</intent-filter>

<intent-filter>
    <action android:name="android.nfc.action.NDEF_DISCOVERED" />
    <category android:name="android.intent.category.DEFAULT" />
    <data android:scheme="http"
          android:host="www.javadude.com"
          android:pathPrefix=""/>
</intent-filter>
```

Simple Read – Process Intent

```
Parcelable[] messages =
      intent.getParcelableArrayExtra(
            NfcAdapter.EXTRA_NDEF_MESSAGES);

if (messages != null) {
  for (Parcelable message : messages) {
    NdefMessage ndefMessage = (NdefMessage) message;
    NdefRecord[] records = ndefMessage.getRecords();
    for (NdefRecord record : records) {
        // process record
    }
}
```

Simple Read – Process Record

```
switch (record.getTnf()) {
  case NdefRecord.TNF_WELL_KNOWN:
    if (Arrays.equals(record.getType(),
                      NdefRecord.RTD_URI)) …
    else if (Arrays.equals(record.getType(),
             NdefRecord.RTD_TEXT)) …
    break;
  case NdefRecord.TNF_ABSOLUTE_URI: … break;
  case NdefRecord.TNF_EMPTY:        … break;
  case NdefRecord.TNF_EXTERNAL_TYPE:
    // new String(record.getId());
    // new String(record.getType());
    // new String(record.getPayload());
    break;
  case NdefRecord.TNF_MIME_MEDIA: … break;
  case NdefRecord.TNF_UNCHANGED:  … break;
  case NdefRecord.TNF_UNKNOWN:    … break;
}
```

NFC – Foreground Read

- Current Application waits to read a tag
- Application is given priority to handle all tags
  - For which it registers intent filters
  - No user request dialog displayed if other apps registered!
- Steps
  - Add NFC permission to manifest
  - Set up intent filters (programmatically)
  - Get NFC Adapter (handle/proxy to NFC service)
  - enableForegroundDispatch (filters + PendingIntent)
  - Catch tag via onNewIntent(Intent)
  - disableForegroundDispatch

NFC – Setup Intent Filters

```
IntentFilter ndefJavaDudeDiscovered =
  new IntentFilter(NfcAdapter.ACTION_NDEF_DISCOVERED);

ndefJavaDudeDiscovered.addDataScheme("http");
ndefJavaDudeDiscovered.addDataAuthority("www.javadude.com",
                                        null);

IntentFilter ndefTextDiscovered =
  new IntentFilter(NfcAdapter.ACTION_NDEF_DISCOVERED,
                   "text/plain");

IntentFilter[] readFilters =
  new IntentFilter[] {
    ndefJavaDudeDiscovered,
    ndefTextDiscovered
  };
```

NFC - Setup Pending Intent

- Describes how to get back to this activity
  - Create Intent that will be used in a startActivity call
    - Should use EXPLICIT Intent!
  - Wrap it in a PendingIntent to capture the current context (to retain permission to call this Activity)

```
Intent togo = new Intent(this, getClass());
togo.addFlags(Intent.FLAG_ACTIVITY_SINGLE_TOP);

PendingIntent pendingIntent =
   PendingIntent.getActivity(
       this, // context
       0,    // request code (unused)
       togo, // intent to launch
       0);   // flags
```

NFC - Enable/Disable Dispatch

```
NfcAdapter nfc =
    NfcAdapter.getDefaultAdapter(this);

nfc.enableForegroundDispatch(
    this,              // context
    pendingIntent,     // intent to send
    readFilters,       // intent filters
    null);             // tech lists

nfc.disableForegroundDispatch(this);
```

NFC – Writing Tags (1)

- Requires NFC permission
- Setup PendingIntent as before
  - Can reuse same PendingIntent for read *and* write

- Create IntentFilter for writing

```
IntentFilter tagFound = new
IntentFilter(
    NfcAdapter.ACTION_TAG_DISCOVERED);
writeFilters = new IntentFilter[] {
tagFound };
```

NFC – Writing Tags (2)

- Setup NDEF message – URI general example:

```
byte[] uri =
    "javadude.com/fee".getBytes(
        Charset.forName("US-ASCII"));

byte[] payload = new byte[uri.length + 1];
payload[0] = 0x01;
   // prefix: 01 = "http://www.",
   //         02 = "https://www.",
   //         03="http://"
   //         ...

System.arraycopy(uri, 0, payload, 1, uri.length);
NdefRecord record =
    new NdefRecord(NdefRecord.TNF_WELL_KNOWN,
                   NdefRecord.RTD_URI,
                   NO_ID, payload);
messageToWrite =
    new NdefMessage(new NdefRecord[] {record});
```

NFC – Writing Tags (2a)

- Recent Android versions contain
  - NdefRecord.*createUri*
  - NdefRecord.*createMime*

- *Nice of them to wrap those up for us…*
  - *But no "create text" as a TNF_WELL_KNOWN!*
  - *Grrrrrr…*

NFC – Writing Tags (3)

- Enable writing
  - Same as enable reading; just pass writeFilters

```
NfcAdapter nfc =
    NfcAdapter.getDefaultAdapter(this);

nfc.enableForegroundDispatch(
    this,
    pendingIntent,
    writeFilters,
    null);
```

NFC Peer-to-Peer

- Can set up one message to send

```
NdefMessage messageToPush =
  new NdefMessage(
    new NdefRecord[] {createText("Foo!")});
nfc.setNdefPushMessage(messageToPush, this);
```

- Or create messages as needed

```
nfc.setNdefPushMessageCallback(
  new CreateNdefMessageCallback() {
    public NdefMessage
           createNdefMessage(NfcEvent event) {
      return new NdefMessage(
        new NdefRecord[] {createText("...")});
    }
  }, this);
```